

Test Evidence — Campus Equipment Booking API

Midterm Practical Lab Test · Platform Development · Thaw Zin Myo Aung (6731503088)

Live API (Base URL)	https://equipment-booking-api.platformdev.workers.dev/api
Quick check	https://equipment-booking-api.platformdev.workers.dev/api/equipment
GitHub repository	https://github.com/Thaw-Zin-Myo-Aung/ThawZinMyoAung-equipment-booking-api
Stack	TypeScript + Hono on Cloudflare Workers, Cloudflare D1 (SQLite), parameter-bound SQL
Test tools	Postman (collection + "Booking API - Cloudflare (live)" environment) and curl (tests/run_tests.sh, based on the cURL Quick Test Guide)

Results summary

Run	Target	Result
Postman collection run (16 screenshots in this document)	Live Cloudflare URL	34 tests, 0 errors
curl suite (guide steps 1–9 + edge cases)	Live Cloudflare URL	26 passed, 0 failed
curl suite — before Quality Gate (v1, commit 04dc9d3)	localhost:8787	16 passed, 4 failed
curl suite — after Quality Gate (v2, commit 2694442)	localhost:8787	26 passed, 0 failed

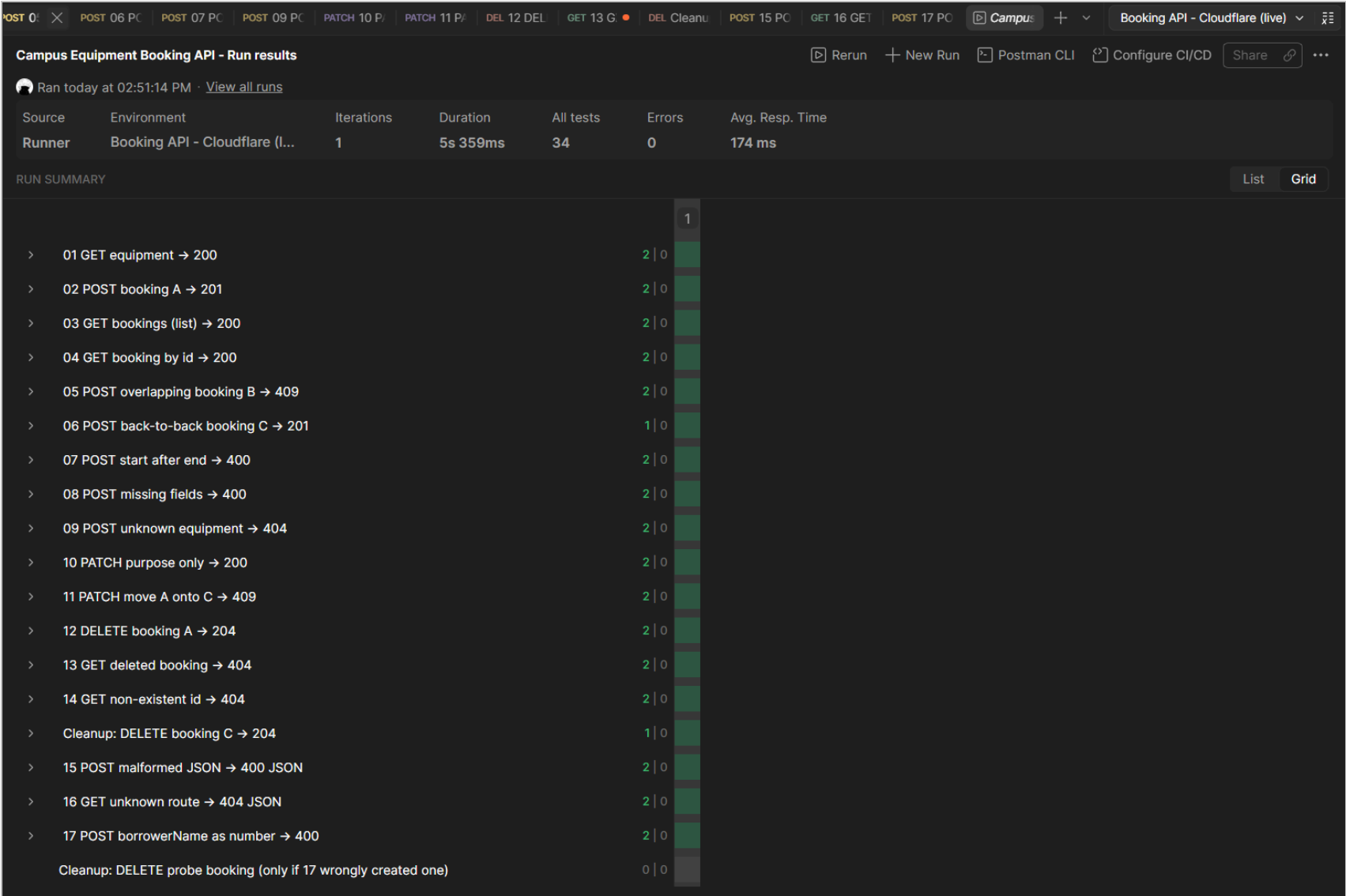
Screenshot index (live Cloudflare URL)

#	Case	Request	Expected	Actual
1	Full Postman collection run	Runner	34 tests	34 / 0 errors
2	List equipment	GET /equipment	200	200
3	Create booking (CREATE)	POST /bookings	201	201
4	Get one booking (READ)	GET /bookings/:id	200	200
5	Overlapping booking on create (CONFLICT)	POST /bookings	409	409
6	Back-to-back booking is allowed	POST /bookings	201	201
7	startAt after endAt (INVALID INPUT)	POST /bookings	400	400
8	equipmentId does not exist	POST /bookings	404	404
9	Partial update (UPDATE)	PATCH /bookings/:id	200	200
10	Overlap on update (CONFLICT)	PATCH /bookings/:id	409	409
11	Update overlapping only its OWN old time	PATCH /bookings/:id	200	200
12	Delete booking (DELETE)	DELETE /bookings/:id	204	204
13	Get deleted booking (NOT FOUND)	GET /bookings/:id	404	404
14	Malformed JSON body (Quality Gate fix)	POST /bookings	400	400
15	Unknown route returns JSON (Quality Gate fix)	GET /nope	404	404
16	Wrong type: borrowerName = 123 (Quality Gate fix)	POST /bookings	400	400

1. Full Postman collection run

Expected 34 tests · Got 34 / 0 errors

Runner · Base URL `https://equipment-booking-api.platformdev.workers.dev/api` · All 19 requests in order on environment "Booking API - Cloudflare (live)". Probe-cleanup is skipped because #17 correctly created nothing.



2. List equipment

Expected 200 · Got 200

GET /equipment · Base URL `https://equipment-booking-api.platformdev.workers.dev/api` · Seeded equipment (3 records) returned from the remote D1 database.

The screenshot shows a REST client interface with the following details:

- Request:**
 - Method: GET
 - URL: `{{baseUrl}}/equipment`
 - Base URL dropdown: `https://equipment-booking-api.platformdev.workers.dev/api`
 - Environment: E Environment
 - Variables in request: →
 - GraphQL: JSON
 - Schema: Schema
 - Beautify: Beautify
- Response:**
 - Status: 200 OK
 - Time: 468 ms
 - Size: 701 B
 - Save Response: Save Response
- Body (JSON):**

```
1 [
2   {
3     "id": "eq-1",
4     "name": "Projector A",
5     "location": "Building 1"
6   },
7   {
8     "id": "eq-2",
9     "name": "Camera Canon EOS",
10    "location": "Media Lab, Building 2"
11  },
12  {
13    "id": "eq-3",
14    "name": "Meeting Room 301",
15    "location": "Building 3"
16  }
17 ]
```

3. Create booking (CREATE)

Expected 201 · Got 201

POST /bookings · Base URL `https://equipment-booking-api.platformdev.workers.dev/api` · Server-generated UUID; times normalised to UTC "Z" format.

The screenshot shows a REST client interface with the following details:

- Request:**
 - Method: POST
 - URL: `{{baseUrl}}/bookings`
 - Body (JSON):

```
{  "equipmentId": "eq-1",  "borrowerName": "Somchai Jaidee",  "startAt": "2026-10-20T09:00:00.000Z",  "endAt": "2026-10-20T11:00:00.000Z",  "purpose": "Class presentation"}
```
- Response:**
 - Status: 201 Created
 - Time: 232 ms
 - Size: 808 B
 - Body (JSON):

```
{  "id": "811b8413-8594-42d8-803b-6bee444eeee0",  "equipmentId": "eq-1",  "borrowerName": "Somchai Jaidee",  "startAt": "2026-10-20T09:00:00.000Z",  "endAt": "2026-10-20T11:00:00.000Z",  "purpose": "Class presentation",  "createdAt": "2026-10-06T07:46:46.849Z"}
```

4. Get one booking (READ)

Expected 200 · Got 200

GET /bookings/:id · Base URL `https://equipment-booking-api.platformdev.workers.dev/api` · Same id as the booking created in 02.

The screenshot shows a REST client interface with the following details:

- Request Method:** GET
- URL:** `{{baseUrl}}/bookings/{{bookingId}}`
- Base URL:** `https://equipment-booking-api.platformdev.workers.dev/api`
- Response Status:** 200 OK
- Response Time:** 561 ms
- Response Size:** 727 B
- Response Body (JSON):**

```
{  "id": "811b8413-8594-42d8-803b-6bee444eeee0",  "equipmentId": "eq-1",  "borrowerName": "Somchai Jaidee",  "startAt": "2026-10-20T09:00:00.000Z",  "endAt": "2026-10-20T11:00:00.000Z",  "purpose": "Class presentation",  "createdAt": "2026-10-06T07:46:46.849Z"}
```

5. Overlapping booking on create (CONFLICT)

Expected 409 · Got 409

POST /bookings · Base URL `https://equipment-booking-api.platformdev.workers.dev/api` · 10:00-12:00 overlaps existing 09:00-11:00 on eq-1.

The screenshot shows a REST client interface with a tab titled "POST 05 POST overlapping booking B → 409". The request is a POST to `{{baseUrl}}/bookings` with the following JSON body:

```
1 {
2   "equipmentId": "eq-1",
3   "borrowerName": "Suda Rakdee",
4   "startAt": "2026-10-20T10:00:00.000Z",
5   "endAt": "2026-10-20T12:00:00.000Z",
6   "purpose": "Workshop"
7 }
```

The response is a 409 Conflict with the following JSON body:

```
1 {
2   "error": "This equipment is already booked for an overlapping time"
3 }
```

The status bar at the bottom indicates "409 Conflict", "516 ms", and "649 B".

6. Back-to-back booking is allowed

Expected 201 · Got 201

POST /bookings · Base URL `https://equipment-booking-api.platformdev.workers.dev/api` · 11:00-12:00 starts exactly when A ends (half-open interval [start, end)).

The screenshot shows a REST client interface with a tab for "POST 06 POST back-to-". The URL is `{{baseUrl}}/bookings`, which has been expanded to show the base URL `https://equipment-booking-api.platformdev.workers.dev/api`. The request body is a JSON object:

```
1 {
2   "equipmentId": "eq-1",
3   "borrowerName": "Suda Rakdee",
4   "startAt": "2026-10-20T11:00:00.000Z",
5   "endAt": "2026-10-20T12:00:00.000Z",
6   "purpose": "Workshop"
7 }
```

The response is shown in the "Body" tab as JSON:

```
1 {
2   "id": "2fbbe5ba-8280-412c-b52e-0b0ff01b3d9b",
3   "equipmentId": "eq-1",
4   "borrowerName": "Suda Rakdee",
5   "startAt": "2026-10-20T11:00:00.000Z",
6   "endAt": "2026-10-20T12:00:00.000Z",
7   "purpose": "Workshop",
8   "createdAt": "2026-10-06T07:47:34.119Z"
9 }
```

The status bar at the bottom indicates "201 Created" with a response time of 199 ms and a body size of 799 B.

7. startAt after endAt (INVALID INPUT)

Expected 400 · Got 400

POST /bookings · Base URL `https://equipment-booking-api.platformdev.workers.dev/api` · Rule: startAt must be before endAt.

The screenshot shows a REST client interface for the "Booking API - Cloudflare (live)". The selected test is "07 POST start after end → 400". The request is a POST to `{{baseUrl}}/bookings` with the following JSON body:

```
1 {
2   "equipmentId": "eq-1",
3   "borrowerName": "Somchai Jaidee",
4   "startAt": "2026-10-21T12:00:00.000Z",
5   "endAt": "2026-10-21T10:00:00.000Z",
6   "purpose": "Invalid range"
7 }
```

The response is a **400 Bad Request** with a status of 89 ms and 605 B. The response body is JSON:

```
1 {
2   "error": "startAt must be before endAt"
3 }
```

Buttons for "Fix booking dates", "Fix", and "Decline" are visible. The "Test Results" tab shows 2/2 tests passed.

8. equipmentId does not exist

Expected 404 · Got 404

POST /bookings · Base URL `https://equipment-booking-api.platformdev.workers.dev/api` · Well-formed request referring to equipment that does not exist.

The screenshot shows a REST client interface for the "Booking API - Cloudflare (live)". The selected test is "09 POST unknown equipment → 404". The request is a POST to `{{baseUrl}}/bookings` with the following JSON body:

```
1 {
2   "equipmentId": "eq-999",
3   "borrowerName": "Somchai Jaidee",
4   "startAt": "2026-10-22T09:00:00.000Z",
5   "endAt": "2026-10-22T10:00:00.000Z",
6   "purpose": "Ghost equipment"
7 }
```

The response status is "404 Not Found" (156 ms, 610 B). The response body is JSON:

```
1 {
2   "error": "Equipment 'eq-999' does not exist"
3 }
```

Buttons for "Diagnose" and "Decline" are visible. The interface also includes tabs for Overview, Docs, Params, Environment, Headers, Test Results, and Cookies.

9. Partial update (UPDATE)

Expected 200 · Got 200

PATCH /bookings/:id · Base URL `https://equipment-booking-api.platformdev.workers.dev/api` · Only purpose changed; times and other fields unchanged.

The screenshot shows a REST client interface for a "Booking API - Cloudflare (live)". The breadcrumb path is "Campus Equipment Booking API > 1. Main tests (contract) > 10 PATCH purpose only → 200".

Request:

- Method: PATCH
- URL: `{{baseUrl}}/bookings/{{bookingId}}`
- Base URL: `https://equipment-booking-api.platformdev.workers.dev/api`
- Body (JSON):

```
{
  "purpose": "Updated: final project demo"
}
```

Response:

- Status: 200 OK
- Time: 237 ms
- Size: 739 B
- Body (JSON):

```
{
  "id": "811b8413-8594-42d8-803b-6bee444eeee0",
  "equipmentId": "eq-1",
  "borrowerName": "Somchai Jaidee",
  "startAt": "2026-10-20T09:00:00.000Z",
  "endAt": "2026-10-20T11:00:00.000Z",
  "purpose": "Updated: final project demo",
  "createdAt": "2026-10-06T07:46:46.849Z"
}
```

10. Overlap on update (CONFLICT)

Expected 409 · Got 409

PATCH /bookings/:id · Base URL https://equipment-booking-api.platformdev.workers.dev/api · Moving A to 10:30-11:30 overlaps booking C (11:00-12:00).

OverviewGET 01 GETGET 04 GET bPOST 02 POSGET 03 GET bPOST 05 POSPOST 06 POSPOST 07 POSPOST 09 POSPATCH 10 PATPATCH 11 PATC+Booking API - Cloudflare (live)

Campus Equipment Booking API > 1. Main tests (contract) > 11 PATCH move A onto C → 409SaveShare

PATCH{{baseUrl}}/bookings/ {{bookingId}}Send

DocsParamshttps://equipment-booking-api.platformdev.workers.dev/api

noneform-cE EnvironmentVariables in request →GraphQLJSON

SchemaBeautify

```
1 {
2   "startAt": "2026-10-20T10:30:00.000Z",
3   "endAt": "2026-10-20T11:30:00.000Z"
4 }
```

BodyCookiesHeaders 9Test Results 2/2Thinking...

409 Conflict · 278 ms · 635 B · Save Response

Cancel

JSONPreviewDebug with AI

```
1 {
2   "error": "This equipment is already booked for an overlapping time"
3 }
```

11. Update overlapping only its OWN old time

Expected 200 · Got 200

PATCH /bookings/:id · Base URL `https://equipment-booking-api.platformdev.workers.dev/api` · Run after C was deleted: the booking being updated is excluded (id <> ?), so no false conflict. (The request's scripted tests target the 409 case, hence 0/2.)

The screenshot shows a REST client interface for the "Booking API - Cloudflare (live)". The selected method is PATCH, and the URL is `{{baseUrl}}/bookings/{{bookingId}}`. The base URL is `https://equipment-booking-api.platformdev.workers.dev/api`. The request body is a JSON object:

```
1 {
2   "startAt": "2026-10-20T10:30:00.000Z",
3   "endAt": "2026-10-20T11:30:00.000Z"
4 }
```

The response status is 200 OK, with a response time of 263 ms and a body size of 733 B. The response body is a JSON object:

```
1 {
2   "id": "9e4ec9d3-85df-4320-9f37-7f1759aa26ab",
3   "equipmentId": "eq-1",
4   "borrowerName": "Somchai Jaidee",
5   "startAt": "2026-10-20T10:30:00.000Z",
6   "endAt": "2026-10-20T11:30:00.000Z",
7   "purpose": "Class presentation",
8   "createdAt": "2026-10-06T07:50:28.527Z"
9 }
```

12. Delete booking (DELETE)

Expected 204 · Got 204

DELETE /bookings/:id · Base URL https://equipment-booking-api.platformdev.workers.dev/api · Empty response body.

OverviewGET 01 GET 04 GET 02 POST 03 GET 05 POST 06 POST 07 POST 09 POST 10 PATCH 11 PATCH 12 DELETE

Campus Equipment Booking API > 1. Main tests (contract) > 12 DELETE booking A → 204

SaveShare

DELETE{{baseUrl}}/bookings/{{bookingId}}Send

DocsParamsAuthorizationHeaders 6BodyScriptsSettingsCookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

BodyCookiesHeaders 7Test Results 2/2204 No Content · 151 ms · 517 B · Save Response

RawPreviewVisualize

1

13. Get deleted booking (NOT FOUND)

Expected 404 · Got 404

GET /bookings/:id · Base URL https://equipment-booking-api.platformdev.workers.dev/api · JSON error {"error":"Booking not found"}.

OverviewGET 01 GET 04 GET 02 POST 03 GET 05 POST 06 POST 07 POST 09 PATCH 10 PATCH 11 DEL 12 GET 13 GET 14

Campus Equipment Booking API > 1. Main tests (contract) > 13 GET deleted booking → 404

GET{{baseUrl}} /bookings/ {{bookingId}}

Send

DocsParamsAuthorizationHeaders 6BodyScriptsSettings

noneform-data x-www-form-urlencodedrawbinaryGraphQLJSON

SchemaBeautify

1

BodyCookiesHeaders 9Test Results 2/2

404 Not Found · 148 ms · 590 B · Save Response

Investigate 404 errorDiagnoseDecline

JSONPreviewDebug with AI

```
1 {
2   "error": "Booking not found"
3 }
```

14. Malformed JSON body (Quality Gate fix)

Expected 400 · Got 400

POST /bookings · Base URL `https://equipment-booking-api.platformdev.workers.dev/api` · v1 returned plain-text 500; v2 returns JSON 400.

The screenshot shows a REST client interface for a "Booking API - Cloudflare (live)". The active tab is "POST 15 PC" with the URL `{{baseUri}}/bookings`. The base URL is expanded to `https://equipment-booking-api.platformdev.workers.dev/api`. The request body is set to "JSON" and contains a single line: `{bad`. The response status is "400 Bad Request" (80 ms, 616 B). The response body is JSON:

```
{  "error": "Request body must be valid JSON"}
```

. The interface includes tabs for Body, Cookies, Headers, and Test Results, as well as buttons for Send, Save, and Beautify.

15. Unknown route returns JSON (Quality Gate fix)

Expected 404 · Got 404

GET /nope · Base URL https://equipment-booking-api.platformdev.workers.dev/api · v1 returned plain-text "404 Not Found"; v2 returns JSON.

POST 02 PCGET 03 GETPOST 05 PCPOST 06 PCPOST 07 PCPOST 09 PCPATCH 10 PPATCH 11 PDEL 12 DELGET 13 GDEL CleanuPOST 15 PCGET 16 GET+Booking API - Cloudflare (live)

Campus Equipment Booking API > 2. Quality Gate probes (expected to FAIL on v1) > 16 GET unknown route → 404 JSONSaveShare

GET{{baseUrl}} /nopeSend

DocsParamsAuthorizationHeaders 6BodyScriptsSettingsCookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

BodyCookiesHeaders 9Test Results 2/2404 Not Found · 81 ms · 608 B · Save Response

Thinking ..Cancel

{ } JSONPreviewDebug with AI

1 {
2 "error": "Route GET /api/nope not found"
3 }

16. Wrong type: borrowerName = 123 (Quality Gate fix)

Expected 400 · Got 400

POST /bookings · Base URL `https://equipment-booking-api.platformdev.workers.dev/api` · v1 accepted and stored a number (201); v2 rejects it.

The screenshot shows a REST client interface for a "Booking API - Cloudflare (live)". The selected endpoint is `POST {{baseUrl}}/bookings`. The request body is a JSON object:

```
1 {
2   "equipmentId": "eq-2",
3   "borrowerName": 123,
4   "startAt": "2026-10-23T09:00:00.000Z",
5   "endAt": "2026-10-23T10:00:00.000Z",
6   "purpose": "Type check"
7 }
```

The response is a **400 Bad Request** with a status of 400, a response time of 106 ms, and a body size of 624 B. The response body is a JSON object:

```
1 {
2   "error": "borrowerName must be a non-empty string"
3 }
```

Below the response body, there is a message: "Add borrowerName to request body".

curl test run against the live Cloudflare URL — 26 passed, 0 failed

Command: `BASE_URL=https://equipment-booking-api.platformdev.workers.dev/api bash tests/run_tests.sh evidence/deployed_cloudflare.txt` · Run at 2026-10-06T07:38:17Z · Full request/response output (curl -i) in `evidence/deployed_cloudflare.txt` . Each error case is also checked to have a JSON `{"error": "..."} body`.

Case	Expected	Got	Result
1. List equipment	200	200	PASS
2. List bookings	200	200	PASS
3. Create a booking	201	201	PASS
4. Get one booking	200	200	PASS
5. Update a booking	200	200	PASS
6. Invalid time range	400	400	PASS
7. Overlapping booking	409	409	PASS
8. Missing booking	404	404	PASS
E1. Back-to-back booking (14:00 start = previous end) is allowed	201	201	PASS
E2. PATCH that moves a booking onto another booking	409	409	PASS
E3. Same time, different equipment is allowed	201	201	PASS
E4. Unknown equipmentId	404	404	PASS
E5. Missing required fields	400	400	PASS
E6. Malformed JSON body	400	400	PASS
E7. Wrong type (borrowerName is a number)	400	400	PASS
E8. Non-ISO date format	400	400	PASS
E9. Unknown route returns JSON	404	404	PASS
E11. Impossible calendar date (Feb 30)	400	400	PASS
E12. Whitespace-only borrowerName	400	400	PASS
E13. PATCH with no updatable fields	400	400	PASS
E14. PATCH only startAt, after the existing endAt (merged range invalid)	400	400	PASS
E15. PATCH a booking that does not exist	404	404	PASS
E16. DELETE a booking that does not exist	404	404	PASS

Case	Expected	Got	Result
E10. SQL injection attempt is treated as plain data	404	404	PASS
9. Delete a booking	204	204	PASS
9b. Get deleted booking	404	404	PASS

Quality Gate: before (v1) vs after (v2)

Same curl suite on the local server. Before: **16 passed, 4 failed** (evidence/before_v1.txt) → After: **26 passed, 0 failed** (evidence/after_v2.txt). Details of each fix: `QUALITY_GATE_REVIEW.md` .

Case that failed on v1	Expected	v1	v2
E6. Malformed JSON body	400	500 (body is NOT JSON {error})	400 PASS
E7. Wrong type (borrowerName is a number)	400	201 (body is NOT JSON {error})	400 PASS
E8. Non-ISO date format	400	201 (body is NOT JSON {error})	400 PASS
E9. Unknown route returns JSON	404	404 (body is NOT JSON {error})	404 PASS